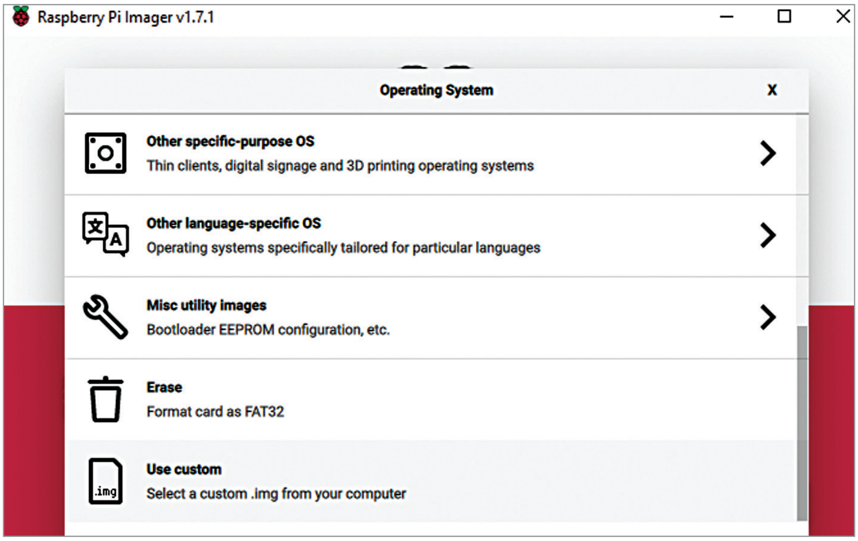


Fig. 10: Preparing SD card with Kali Linux OS

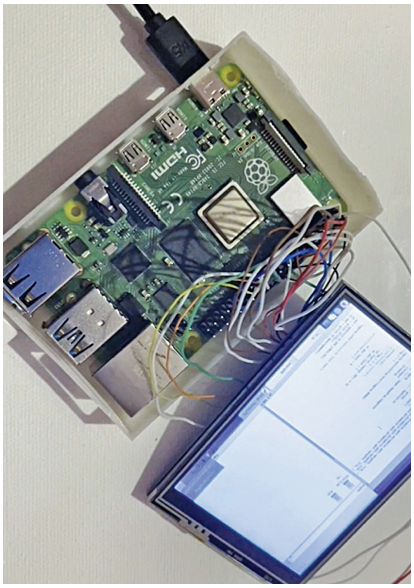

Fig. 11: Connecting SPI display with Raspberry Pi

stead. By using the HDMI display, you can skip the following steps (required for SPI display) and go straight to the keyboard preparation section.

To use the SPI display, we need to install the drivers and setup the SPI touch display. So, open the terminal and run the following command in Linux terminal to install the driver and setup the display for Raspberry Pi:

```
git clone https://github.com/goodtft/
LCD-show.git
chmod -R 755 LCD-show
cd LCD-show/
sudo ./LCD35-show
```

After installing the driver, connect the SPI display, as shown in Fig. 11, and solder the pins to Raspberry Pi as per Table 2.

## Preparing the touch keyboard

We need to prepare the keypad system for the laptop for giving input to the laptop. As we wish to make a full touch keyboard panel, we are using the touch e-ink display as the touch based keypad, but we need software to use it as keyboard.

For making the touch keyboard, we need to display the keys on the e-ink display. So, download a vector image having the keys on the keyboard. Resize the image to the size

Fig. 12: Keyboard layout on e-ink display

**TABLE 2**
**SPI DISPLAY CONNECTIONS**

| Pin No. | Symbol | Description |
|---|---|---|
| 1, 17 | 3.3V | Power positive (3.3V power input) |
| 2, 4 | 5V | Power positive (5V power input) |
| 3, 5, 7, 8, 10, 12, 13, 15, 16 | NC | NC |
| 6, 9, 14, 20, 25 | GND | Ground |
| 11 | TP_IRQ | Touch panel interrupt; low level while the touch panel detects touching |
| 18 | LCD_RS | Instruction/data register selection |
| 19 | LCD_SI / TP_SI | SPI data input of LCD/touch panel |
| 21 | TP_SO | SPI data output of touch panel |
| 22 | RST | Reset |
| 23 | LCD_SCK / TP_SCK | SPI clock of LCD/touch panel |
| 24 | LCD_CS | LCD chip selection, low active |
| 26 | TP_CS | Touch panel chip selection, low active |